

2024 Calculus II Additional Exercise (9)

1. Exercise 13, 14, 15, 17, 20 in page 40 from Calculus Supplement Homework.
2. Let the curve L be the intersecting line of surfaces $z = x^2 + y^2$ and $2x^2 + 2y^2 - z^2 = 0$. Find parametric equations for the line tangent to L at the point $P_0(1, 1, 2)$.
3. Find the local extreme values of the function $f(x, y) = 2(y - x^2)^2 - \frac{1}{7}x^7 - y^2$.
4. Exercise 27 in page 41 from Calculus Supplement Homework.
5. Exercise 29 in page 41 from Calculus Supplement Homework.

Q1 P40 Ex 13, 14, 15, 17, 20

13. Assume the tangent plane of $xyz = 6$ at the point P is parallel to $6x - 3y + 2z + 1 = 0$. The position of P is ().
14. The absolute minimum of $z = x^2 + y^2$ with the restriction $x + y = 1$ is ().
15. $u = x^{y^z}$, $x > 0$, $y > 0$, then $du = ()$.
16. Let $z = z(x, y)$ be the function determined implicitly by $e^{-xy} - 2z + e^{xz} = 0$. Then $\left. \frac{\partial^2 z}{\partial x^2} \right|_{(0,1)} = ()$.
17. Suppose the tangent plane of the surface $z - e^z + 2xy = 3$ at P is $x + 2y = C$. Then $P = ()$ and $C = ()$.
18. ★ The minimum of the distance between a point on $y = x^2$ and a point on $x - y - 2 = 0$ is ().
19. Let $w = g(x, y) = f(x, y, z(x, y)) = e^x y z^2$, here $z(x, y)$ is determined by $x + y + z + xyz = 2$. Then $\left. \frac{\partial w}{\partial x} \right|_{(0,1)} = ()$.
20. Assume $f(u)$ is differentiable and $f'(0) = 0.5$. Then $z = f(4x^2 - y^2)$ at $(1, 2)$ increases fastest in the direction of ().

Sol to Ex 13:

$$F(x, y, z) = xyz - 6$$

$$\left(\frac{\partial F}{\partial x} \quad \frac{\partial F}{\partial y} \quad \frac{\partial F}{\partial z} \right) = (yz \quad xz \quad xy) // (6, -3, 2)$$

$$\Rightarrow \begin{cases} \frac{yz}{6} = \frac{xz}{-3} \\ \frac{yz}{6} = \frac{xy}{2} \end{cases} \xrightarrow{\text{解得}} \begin{cases} y = -2x \\ z = 3x \end{cases}$$

$$\text{回代到 } F(x, y, z) = 0 \text{ 得 } -6x^3 - 6 = 0$$

$$\text{解得 } \begin{cases} x = -1 \\ y = 2 \\ z = 3 \end{cases} \quad \text{即 } P = (-1, 2, 3)$$

□

Sol to Ex14:

(法一): 根据拉格朗日乘子法: 设 $L(x, y, \lambda) = x^2 + y^2 - \lambda(x + y - 1)$

$$\nabla L = 0 \Leftrightarrow \begin{cases} L'_x = 0 \\ L'_y = 0 \\ L'_\lambda = 0 \end{cases} \Leftrightarrow \begin{cases} 2x - \lambda = 0 \\ 2y - \lambda = 0 \\ x + y - 1 = 0 \end{cases} \xrightarrow{\text{解得}} \begin{cases} x = \frac{\lambda}{2} = \frac{1}{2} \\ y = \frac{\lambda}{2} = \frac{1}{2} \\ \lambda = 1 \end{cases}$$

$\therefore$ $x = y = \frac{1}{2}$ 时, $z = x^2 + y^2 = \frac{1}{2}$ 为 $\{(x, y) | x + y = 1\}$ 上的绝对最大值或绝对最小值.

$\therefore$ $x = 1, y = 0$ 时, $z = 1 + 0 = 1 > \frac{1}{2}$

$\therefore$ 上述值 $\frac{1}{2}$ 为最小值.

(法二) 根据基本不等式: $z = x^2 + y^2 \geq \frac{(x+y)^2}{2} = \frac{1}{2}$

等式成立当且仅当 $x = y = \frac{1}{2}$.

□

Sol to Ex15:

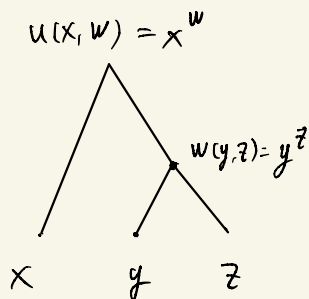
(法一)

$$\frac{\partial u}{\partial x} = w x^{w-1} = \underline{x^{(y^z-1)} y^z}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial w} \cdot \frac{\partial w}{\partial y} = x^w \cdot \ln x \cdot z \cdot y^{z-1} = \underline{x^{(y^z)} \cdot \ln x \cdot z \cdot y^{z-1}}$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial w} \cdot \frac{\partial w}{\partial z} = x^w \cdot \ln x \cdot y^z \cdot \ln y = \underline{x^{(y^z)} \cdot \ln x \cdot \ln y \cdot y^z}$$

$$\text{Then } du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz.$$



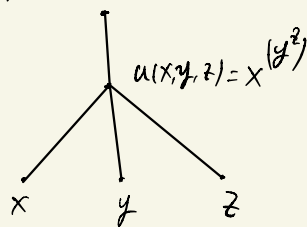
(法=)

$$\ln u = \ln x^{(y^z)} = y^z \cdot \ln x$$

$$v := \ln \ln u = \ln(y^z \cdot \ln x) = z \ln y + \ln \ln x$$

$$\text{Then } \frac{dv}{du} = \frac{1}{u \cdot \ln u}$$

$$v(u) = \ln \ln u$$



$$\left\{ \begin{array}{l} \frac{\partial v}{\partial x} = \frac{dv}{du} \cdot \frac{\partial u}{\partial x} \\ \frac{\partial v}{\partial y} = \frac{dv}{du} \cdot \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial z} = \frac{dv}{du} \cdot \frac{\partial u}{\partial z} \end{array} \right. \begin{array}{l} \text{把 } \frac{dv}{du} \\ \text{除到} \\ \Rightarrow \\ \text{左边去} \end{array} \left\{ \begin{array}{l} \frac{\partial u}{\partial x} = u \cdot \ln u \cdot \frac{1}{x \ln x} = x^{(y^z)} \cdot y^z \ln x \cdot \frac{1}{x \ln x} \\ \quad = x^{(y^z-1)} \cdot y^z \\ \frac{\partial u}{\partial y} = u \cdot \ln u \cdot \frac{z}{y} = x^{(y^z)} \cdot \ln x \cdot y^{z-1} \cdot z \\ \frac{\partial u}{\partial z} = u \cdot \ln u \cdot \ln y = x^{(y^z)} \cdot y^z \cdot \ln x \cdot \ln y \end{array} \right.$$

□

Sol to Ex17:

$$\text{令 } F(x, y, z) = z - e^z + 2xy - 3$$

$$\nabla F = \frac{\partial F}{\partial x} \vec{i} + \frac{\partial F}{\partial y} \vec{j} + \frac{\partial F}{\partial z} \vec{k} = 2y \vec{i} + 2x \vec{j} + (1 - e^z) \vec{k}$$

$$\text{P点处切平面: } x + 2y - C = 0, \text{ 那么法向量 } \vec{n} = (1, 2, 0)$$

$$\therefore \nabla F \parallel \vec{n}$$

$$\therefore \begin{cases} 2 \cdot 2y = 1 \cdot 2x \\ 1 - e^z = 0 \end{cases} \text{ 解得 } \begin{cases} x = 2y \\ z = 0 \end{cases}$$

$$\text{代入曲面 } F(x, y, z) = 0: 0 - e^0 + 4y^2 - 3 = 0 \text{ 解得 } y = \pm 1$$

$$\therefore \text{P点坐标 } (2, 1, 0) \text{ 或 } (-2, 1, 0)$$

$$\text{代入切平面解得 } C = 2 + 2 = 4 \text{ 或 } C = -2 - 2 = -4.$$

□

Sol to Ex 20:

$$\begin{cases} \frac{\partial z}{\partial x} = f' \cdot 8x \\ \frac{\partial z}{\partial y} = f' \cdot (-2y) \end{cases} \Rightarrow \begin{cases} \frac{\partial z}{\partial x} \big|_{(1,2)} = f'(0) \cdot 8 = 4 \\ \frac{\partial z}{\partial y} \big|_{(1,2)} = f'(0) \cdot (-4) = -2 \end{cases}$$

$$\therefore \text{梯度 } \nabla z \big|_{(1,2)} = 4\vec{i} - 2\vec{j}$$

$$\vec{u} \text{ 方向导数 } \left(\frac{dz}{ds} \right)_u \big|_{(1,2)} = \nabla z \big|_{(1,2)} \cdot \vec{u} \quad \text{maximal iff } \vec{u} \parallel \nabla z \big|_{(1,2)}$$

$$\therefore \vec{u} = \frac{1}{\sqrt{5}}(2\vec{i} - \vec{j})$$

□

Q2

2. Let the curve L be the intersecting line of surfaces $z = x^2 + y^2$ and $2x^2 + 2y^2 - z^2 = 0$. Find parametric equations for the line tangent to L at the point $P_0(1, 1, 2)$.

Sol: 计算 P_0 处两个曲面切空间的交线.

$$S_1 := \{(x, y, z) \mid x^2 + y^2 - z = 0\}$$

$$S_2 := \{(x, y, z) \mid 2x^2 + 2y^2 - z^2 = 0\}$$

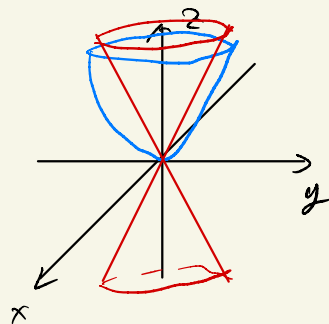
$$\vec{n}_1|_{P_0} = (2, 2, 1)$$

$$T_1 = \{(x, y, z) \mid 2(x-1) + 2(y-1) - (z-2) = 0\}$$

$$\vec{n}_2|_{P_0} = (4, 4, -4)$$

$$T_2 = \{(x, y, z) \mid x-1 + y-1 - (z-2) = 0\}$$

$$\begin{aligned} \text{交线: } \begin{cases} 2x + 2y - z - 2 = 0 \\ x + y - z = 0 \end{cases} &\Rightarrow \begin{cases} x + y - 2 = 0 \\ z = 2 \end{cases} \Rightarrow \begin{cases} x + 2 = -t \\ y = t \\ z - 2 = 0 \end{cases} \end{aligned}$$



□

Q3

3. Find the local extreme values of the function $f(x, y) = 2(y-x^2)^2 - \frac{1}{7}x^7 - y^2$.

Sol. $\frac{\partial f}{\partial x} = 4(y-x^2) \cdot (-2x) - x^6 = -8xy + 8x^3 - x^6$

$$\frac{\partial f}{\partial y} = 4(y-x^2) - 2y = 2y - 4x^2$$

$$\frac{\partial^2 f}{\partial x^2} = -8y + 24x^2 - 6x^5$$

$$\frac{\partial^2 f}{\partial x^2} \Big|_{(-2, 8)} = -8^2 + 3 \cdot 8 \cdot 4 + 3 \cdot 2^6 = 32 + 288 = 160$$

$$\frac{\partial^2 f}{\partial y \partial x} = -8x$$

$$\frac{\partial^2 f}{\partial x \partial y} \Big|_{(-2, 8)} = 16$$

$$\frac{\partial^2 f}{\partial x \partial y} = -8x$$

$$\frac{\partial^2 f}{\partial y^2} = 2$$

$$\begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases} \Rightarrow -8x^3 - x^6 = 0 \text{ 解得 } \begin{cases} x=0 \\ y=0 \end{cases} \text{ 或 } \begin{cases} x=-2 \\ y=8 \end{cases}$$

$$\text{Hess } f \Big|_{(0,0)} = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\text{Hess } f \Big|_{(-2,8)} = \begin{bmatrix} 160 & 16 \\ 16 & 2 \end{bmatrix}$$

$$\therefore \begin{vmatrix} 160 & 16 \\ 16 & 2 \end{vmatrix} > 0 \text{ and } f_{xx} \Big|_{(-2,8)} = 160 > 0$$

$$\Rightarrow f(-2, 8) = -\frac{96}{7} \text{ is local minimum value}$$

But $(0, 0)$ is not local extreme point since

$$\textcircled{1} f(x, y) \Big|_{\{y=0\}} = 2x^4 - \frac{1}{7}x^7 = x^4 \left(2 - \frac{1}{7}x^3 \right)$$

Then we can find a sequence of points $(x_n, 0)$ with
 $f(x_n, 0) = x_n^4 (2 - \frac{1}{7}x_n^3) > 0$ and $\lim_{n \rightarrow \infty} x_n = 0$

$$\textcircled{2} \quad f(x, y) \Big|_{\{y=x^2\}} = -\frac{1}{7}x^7 - x^4$$

Then we can find a sequence of points $(\tilde{x}_n, \tilde{x}_n^2)$ with
 $f(\tilde{x}_n, \tilde{x}_n^2) = -\frac{1}{7}\tilde{x}_n^7 - \tilde{x}_n^4 < 0$ and $\lim_{n \rightarrow \infty} \tilde{x}_n = 0$.

□

Q4. P41 Ex27

27. Find the absolute maximum and minimum for $f(x, y) = x^2 + 2y^2 - x^2y^2$ on $D = \{(x, y) \mid x^2 + y^2 \leq 4, y \geq 0\}$.

Sol.

$$f(x, y) = x^2 + 2y^2 - x^2y^2$$

$$\frac{\partial f}{\partial x} = 2x - 2xy^2$$

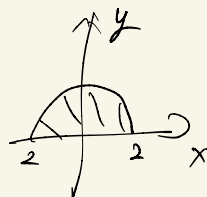
$$\frac{\partial f}{\partial y} = 4y - 2x^2y$$

$$\frac{\partial^2 f}{\partial x^2} = 2 - 2y^2$$

$$\frac{\partial^2 f}{\partial x \partial y} = -4xy$$

$$\frac{\partial^2 f}{\partial y \partial x} = -4xy$$

$$\frac{\partial^2 f}{\partial y^2} = 4 - 2x^2$$



$$\begin{cases} \frac{\partial f}{\partial x} = 2x - 2xy^2 = 0 \\ \frac{\partial f}{\partial y} = 4y - 2x^2y = 0 \end{cases} \Rightarrow \begin{cases} x=0 \\ y=0 \end{cases} \text{ or } \begin{cases} x = \pm\sqrt{2} \\ y=1 \end{cases}$$

Lagrange Multipliers on boundary $\{(x, y) \mid x^2 + y^2 = 4\}$

$$L(x, y, \lambda) = f(x, y) - \lambda(x^2 + y^2 - 4)$$

$$\begin{cases} \frac{\partial L}{\partial x} = 2x - 2xy^2 - 2\lambda x = 0 \\ \frac{\partial L}{\partial y} = 4y - 2x^2y - 2\lambda y = 0 \\ \frac{\partial L}{\partial \lambda} = x^2 + y^2 - 4 = 0 \end{cases} \Rightarrow \begin{cases} x = \pm\sqrt{\frac{5}{2}} \\ y = \sqrt{\frac{3}{2}} \\ \lambda = -\frac{1}{2} \end{cases} \text{ or } \begin{cases} x=0 \\ y=2 \\ \lambda=2 \end{cases}$$

$$\begin{aligned} f(0, 0) &= 0 \\ f(\pm\sqrt{2}, 1) &= 1 \end{aligned} \quad \text{or} \quad \begin{cases} x = \pm 2 \\ y = 0 \\ \lambda = 1 \end{cases}$$

$$f\left(\pm\sqrt{\frac{5}{2}}, \sqrt{\frac{3}{2}}\right) = \frac{5}{2} + 2 \cdot \frac{3}{2} - \frac{5}{2} \cdot \frac{3}{2} = \frac{11}{2} - \frac{15}{4} = \frac{7}{4}$$

$$f(0, 2) = 8$$

$$f(\pm 2, 0) = 4$$

∴ maximum 8

minimum 0

□

Q5 P41 Ex 29

29. Assume that $\mathbf{n}$ is the normal unit vector pointing inside at $(1, 1, 1)$ for the surface $2x^2 + 3y^2 + z^2 = 6$. Compute the directional derivative for $u = \frac{\sqrt{6x^2 + 8y^2}}{z}$ at the point in the direction of $\mathbf{n}$.

Sol:

$$F(x, y, z) = 2x^2 + 3y^2 + z^2 - 6 = 0$$

$$\left(\frac{\partial F}{\partial x} \quad \frac{\partial F}{\partial y} \quad \frac{\partial F}{\partial z} \right)^T \Big|_{(1,1,1)} = (4x \quad 6y \quad 2z)^T \Big|_{(1,1,1)} = (4 \quad 6 \quad 2)^T$$

$\therefore$ Normal unit vector pointing inside at $(1, 1, 1)$ is

$$\vec{n} = \frac{1}{\sqrt{4^2 + 6^2 + 2^2}} (-4\vec{i} - 6\vec{j} - 2\vec{k}) = \frac{1}{\sqrt{14}} (-2\vec{i} - 3\vec{j} - \vec{k})$$

$$u = \frac{\sqrt{6x^2 + 8y^2}}{z}$$

$$\left\{ \begin{array}{l} \frac{\partial u}{\partial x} = \frac{12x}{2\sqrt{6x^2 + 8y^2} z} \\ \frac{\partial u}{\partial y} = \frac{16y}{2\sqrt{6x^2 + 8y^2} z} \\ \frac{\partial u}{\partial z} = -\frac{\sqrt{6x^2 + 8y^2}}{z^2} \end{array} \right. \Rightarrow \left\{ \begin{array}{l} \frac{\partial u}{\partial x} \Big|_{(1,1,1)} = \frac{6}{\sqrt{14}} \\ \frac{\partial u}{\partial y} \Big|_{(1,1,1)} = \frac{8}{\sqrt{14}} \\ \frac{\partial u}{\partial z} \Big|_{(1,1,1)} = -\sqrt{14} \end{array} \right.$$

$$\begin{aligned} \text{Then } \frac{du}{d\vec{n}} \Big|_{(1,1,1)} &= \nabla u \Big|_{(1,1,1)} \cdot \vec{n} = \left(\frac{6}{\sqrt{14}} \quad \frac{8}{\sqrt{14}} \quad -\sqrt{14} \right) \begin{pmatrix} -\frac{2}{\sqrt{14}} \\ -\frac{3}{\sqrt{14}} \\ -\frac{1}{\sqrt{14}} \end{pmatrix} \\ &= -\frac{6}{7} - \frac{12}{7} + 1 = -\frac{11}{7} \end{aligned}$$

□